LAB–3: SOFTWARE ARCHITECTURE & UML COMPONENT DESIGN

Food Bank Surplus Redistribution Platform

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1. Architecture Selection and Justification

The Food Bank Surplus Redistribution Platform uses a Microservices Architecture. The system is split into an API Gateway plus independent Food Listing & Expiry, Reservation & Verification, and Notification services, each backed by a shared Database component. This separation keeps the safety-critical reservation flow and the high-volume real-time notification flow independently deployable and scalable.

ARCHITECTURE JUSTIFICATION

Architecture Selection: We chose Microservices Architecture for the Food Bank Surplus Redistribution Platform.

Reason 1: NFR001 requires instant, encrypted push notifications to every shelter within a 5 km radius within 10 seconds, even at peak load. Microservices let the Notification Service be scaled and load-tested independently of the listing and reservation traffic, instead of over-provisioning the whole platform for a single spiky workload.

Reason 2: Because surplus food spoils quickly, the Reservation & Verification Service must stay within the 99.5% uptime target (NFR002) regardless of what happens elsewhere. Splitting notification dispatch and expiry cleanup into separate services means a slowdown in either one cannot cascade into and block a shelter's in-progress reservation.

Security Advantage: The API Gateway verifies shelter credentials before any request reaches a backend service, and the encrypted (TLS) push channel is isolated inside the Notification Service, so a compromise of one service does not automatically expose donor or shelter data held by the others or by the Database component.

Performance Benefit: The time-critical Reservation & Verification Service (3-tap completion, code generated within 5 seconds) runs and scales separately from the heavier real-time notification fan-out, so a burst of notification traffic never slows down reservation confirmations.

2. Component Justification

Client Application Component: Touch/mobile-friendly interface used by Donor Partners to post batches and by Shelter Coordinators to browse, filter, and reserve batches in 3 taps or fewer.

API Gateway Component: Single entry point that verifies shelter credentials and routes each request to the correct backend microservice.

Food Listing & Expiry Service: Handles posting of surplus batches with expiry/dietary tags, dietary-tag/distance filtering, and automatic removal of expired batches with donor notification.

Reservation & Verification Service: Reserves a pickup window on an unclaimed batch and generates the unique pickup verification code shown to both parties at handoff.

Notification Service: Dispatches instant, encrypted push notifications to shelters within 5 km of a new listing, and expiry alerts to donors.

Database Component: Persists food batch listings, reservations, verification codes, and shelter/donor registration data.

3. Interface Justification

Client API: Connects the Client Application to the API Gateway for all donor and shelter actions.

Listing API: Lets the API Gateway forward posting, browsing, and filtering requests to the Food Listing & Expiry Service.

Reservation API: Lets the API Gateway forward pickup-window reservation and verification-code requests to the Reservation & Verification Service.

Notify Interface: Used by both the Food Listing & Expiry Service (expiry notices) and the Reservation & Verification Service (confirmations) to request delivery from the Notification Service.

Listing Data Interface: Enables the Food Listing & Expiry Service to store and retrieve batch-related data from the Database component.

Reservation Data Interface: Enables the Reservation & Verification Service to store and retrieve reservation-related data from the Database component.

Push Delivery Interface (TLS): Required interface through which the Notification Service dispatches messages to shelter devices over an encrypted channel, satisfying NFR001.

4. UML Component Diagram

The following UML component diagram represents the selected microservices architecture, six major components, and eight interfaces. The PNG diagram is embedded below for direct inclusion in the report.

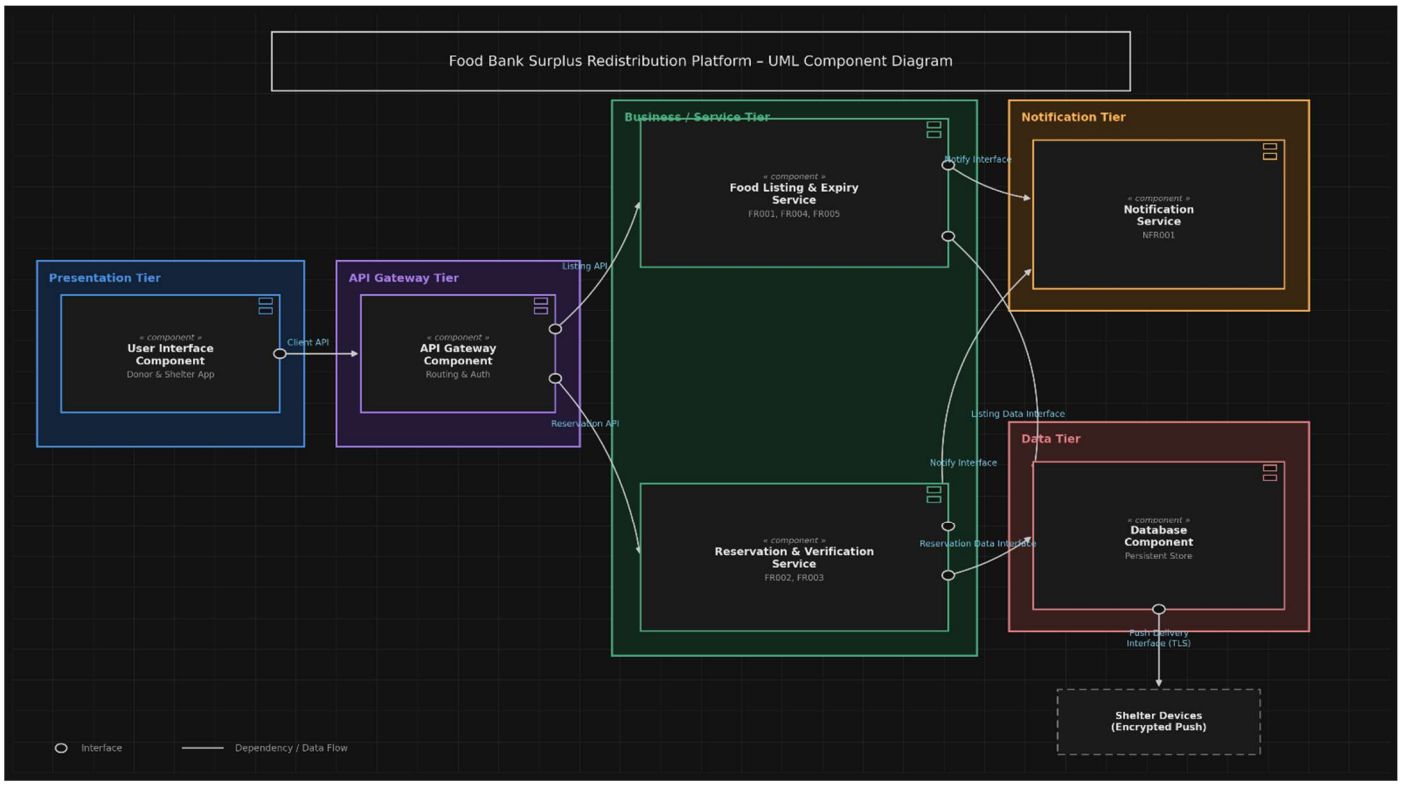


Figure 1: UML Component Diagram – Food Bank Surplus Redistribution Platform

5. Data and Interaction Flow

The Client Application talks only to the API Gateway, which authenticates the request and forwards it to the Food Listing & Expiry Service or the Reservation & Verification Service. Both business services persist their data through the Database component and call the Notification Service whenever a shelter or donor needs to be alerted; the Notification Service then pushes the encrypted message to shelter devices. This keeps the reservation and listing flows decoupled from notification delivery while still sharing a single source of truth for batch and reservation data.

6. Conclusion

The proposed microservices architecture cleanly separates the safety-critical reservation flow from the high-volume, latency-sensitive notification flow. The six components and eight interfaces support independent scaling and fault isolation, letting the platform meet NFR001's real-time delivery target and NFR002's uptime/usability targets without one workload degrading another.